

Ekamveer Singh

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RESEARCH INTERESTS

Reliability and robustness of multi-model / multi-agent LLM systems; verification and escalation strategies for reducing hallucination and error in cascaded LLM pipelines; efficient, on-device, and privacy-preserving inference.

EDUCATION

Punjab Engineering College

B.Tech, CSE-AI (CGPA: 9.01) + Minor, Quantum Technology (CGPA: 10.00)

Chandigarh, India

Expected May 2028

PROJECTS

LLM Cascade – Minimum Sufficient Inference | *Ollama, Python, Evaluation Harness* 2026

- Developed a four-layer weak-to-strong inference ladder (gpt-oss-20b, gemma4, nemotron-3-super, nemotron-3-ultra) to study how much LLM inference is necessary for reliable code generation; stronger models receive the failed candidate and objective execution feedback.
- Built a sandboxed evaluation harness over 100 code-generation problems with deterministic test execution, resumable streaming evaluation, token/latency tracking, and execution-gated escalation based on verified failures.
- L1-only achieved 90% accuracy; execution-gated escalation reached 100% while escalating only 8% of problems, all resolved at L2, with mean token use increasing from 693 to 808 per query. ([repo](#))

Gwen – Local Personal AI Assistant | *MLX, MLX-Whisper, SpeechBrain, Ollama, SwiftUI* Apr. 2026 – Present

- Architected a four-tier model routing system (on-device to cloud) with automatic fallback, and a query-routing layer built specifically to ground responses and reduce hallucination.
- Implemented speaker verification (ECAPA-TDNN) and acoustic echo cancellation (WebRTC AEC3) for reliable, full-duplex, barge-in-capable conversation.
- Built a fully local, privacy-first pipeline on Apple Silicon with multimodal I/O (image/PDF/DOCX), semantic memory, and persistent state; all code written and committed independently across 15 progressive stages. ([repo](#))

Automated Fungal Keratitis Screening | *Python, FastAPI, OpenCV, Scikit-learn* July 2026

- Led system architecture for a 3-person team building a computer vision pipeline to screen fungal keratitis from confocal microscopy images, motivated by a clinical need raised by a PGI-MER physician to faculty.
- Built the LLM-based layer translating a stacked ensemble classifier's outputs and patient risk factors into clinician-readable diagnostic summaries; benchmarked across seven image-processing pipelines.

RESEARCH & PUBLICATIONS

Co-author, “Interface-Driven Electrochemical Performance of Hybrid MXenes for High-Performance Supercapacitors: Insights and Challenges” – *Current Opinion in Solid State & Materials Science*, under review. Independently authored the AI methodology section.

TECHNICAL SKILLS

Languages: Python, C++, Swift, SQL

ML / AI: PyTorch, TensorFlow, MLX, MLX-VLM, Scikit-learn, CatBoost, OpenCV

Systems: MLX Swift, SwiftUI, Apple Silicon inference, FastAPI, Ollama

Data: NumPy, Pandas, Matplotlib, SQL (MySQL, PostgreSQL, MongoDB, SQLite)

ACCOMPLISHMENTS

Finalist, Amazon ML Summer School 2026 | Published op-ed, *The Tribune* (2023)